3240 Pset 4

1. Known;

$G = 800 W/m^2$

$L_g = 3 \times 10^{-3} m$

$k_g = 1.5 W/mK$

$L_a = 0.1 \times 10^{-3} m$

$K_a = 1 W/mK$

$R_c'' = 10^{-4} m^2K/W$

10% absorb glass (q''_{top})

$(q''_{si}) = 85\%$

$\eta = 0.553 - 0.001 T_{si}$

$T_\infty = 300 K$

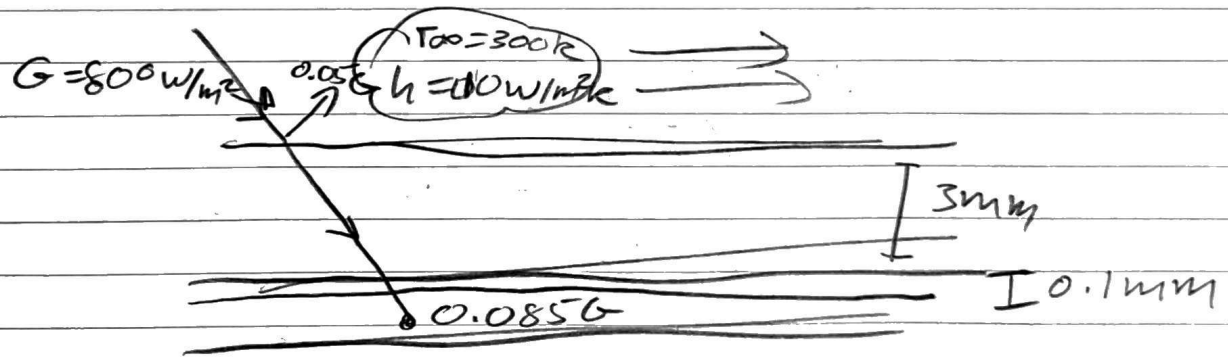
$h = 10 W/m^2K$

$h_{rad} = 5 W/m^2K$

Find: a) Resistance network + values

b) $T_{g,top}$ & T_{si}

c) η & electricity power glass



Assumptions + Properties:

- Steady state
- 1D
- Insulated bottom
- Uniform T_{si}
- Constant thermal conductivities

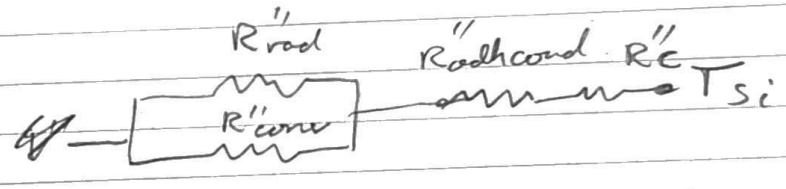
1. a) $R''_{c,\ Si-adhesive} = 10^{-4}\ m^2 \cdot K/W$

$R''_{cond,a} = \frac{L_a}{k_a} = \frac{0.1\times10^{-3}}{1} = 0.1\times10^{-3}\ \frac{m^2 K}{W}$

$R''_{cond,g} = \frac{L_g}{k_g} = \frac{3\times10^{-3}}{1.5} = 2\times10^{-3}\ \frac{m^2 K}{W}$

$R''_{conv} = \frac{1}{h} = \frac{1}{10} = 0.1\ m^2 K/W$

$R''_{rad} = \frac{1}{h_{rad}} = 0.2\ m^2 K/W$



b) $q_{th(Si)} = 0.856\,(1-\eta)$

$= 680\,(1-(0.553-0.001\,T_{Si}))$

$= 303.96 + 0.68\,T_{Si}$

$q''_{g\ absorbed} = 0.1\,G = 80\ W/m^2$

$303.96 + 0.68\,T_{Si} + 80 = 15\,T_{g,top} - 4500$

$\boxed{0.68\,T_{Si} - 15\,T_{g,top} = -4883.96}$

$q''_{th,Si} = \frac{T_{Si} - T_{g,top}}{R''_c + R''_{cond,a} + R''_{cond,g}} = \frac{T_{Si} - T_{g,top}}{0.0022}$

$T_{Si}\left(1 - 0.0022 \times \frac{303.96}{0.68}\right) - T_{g,top} = 0.668712$

$\boxed{0.998504\,T_{Si} - T_{g,top} = 0.668712}$

$\boxed{T_{Si} = 342.29\,K}$ $\boxed{T_{g,top} = 341.11\,K}$

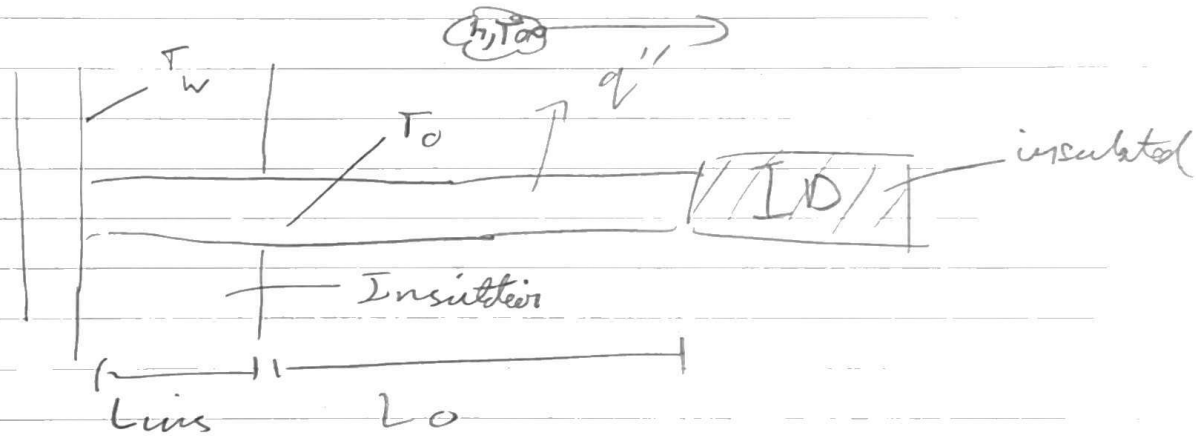
c) $\eta = 0.553 - 0.01(342.29) = 0.21071$

$\boxed{= 21.1\%}$

$P'' = \eta(0.85G)$

$\boxed{= 143.28\ W/m^2}$

2.



Known: $D = 0.01m$ $\quad T_w = 200°C$

$k_{rod} = 80 W/mK$ $\quad L_{ins} = 0.2m$

$T_{max} = 100°C$

$T_\infty = 20°C$

$h = 20 W/m^2K$

Find: a) expression for T_o

b) T_o when $L_o = 200mm$ + check against limit

c) Min L_o for $T_o \leq T_{max}$

Assume:

1) Steady State
2) 1D conduction along rod
3) constant k/h
4) perfect insulation

$m = \sqrt{\frac{hP}{kA_c}}$

$m = 10 m^{-1}$

$$\theta_b = T_b - T_\infty$$
$$= T_o - T_\infty$$

2. a) $q_{cond} = \frac{kA_c}{L_{ins}}(T_w - T_o)$

$q_{fin} = \sqrt{hPkA_c}\,\theta_b \tanh(mL_o)$ $\quad m = \sqrt{\frac{hP}{kA_c}}$

$$q_{cond} = q_{fin}$$

$$\frac{kA_c}{L_{ins}}(T_w - T_o) = \sqrt{hPkA_c}\,(T_o - T_\infty)\tanh(mL_o)$$

$$\frac{1}{L_{ins}}(T_w - T_o) = \frac{\sqrt{hPkA_c}}{kA_c}(T_o - T_\infty)\tanh(mL_o)$$

$$= \sqrt{\frac{hP}{kA_c}} = m$$

$$\frac{T_w - T_o}{L_{ins}} = m(T_o - T_\infty)\tanh(mL_o)$$

$$T_o - T_\infty = \frac{T_w - T_o}{L_{ins}\, m \tanh(mL_o)}$$

$$T_o\left(1 + \frac{1}{L_{ins}\, m \tanh(mL_o)}\right) = \frac{T_w}{L_{ins}\, m \tanh(mL_o)} + T_\infty$$

$$\frac{m L_{ins} \tanh(mL_o) + 1}{m L_{ins} \tanh(mL_o)}$$

$$\boxed{T_o = \frac{T_w + m L_{ins} T_\infty \tanh(mL_o)}{1 + m L_{ins} \tanh(mL_o)}}$$

2.b) $L_o = 0.2m$

$L_{lim} = 0.2m$

$mL_o = 2$

$mL_{lim} = 2$

Sub in:

$T_o = 81.47°C$

$T_o < 100°C$ so within limit

c) let $T_o = T_{max} = 100°C$

plug in: $100 = \frac{200 + 2(\tanh(10L_o))(20)}{1 + 2\tanh(10L_o)}$

$160\tanh(10L_o) = 100$

$10L_o = 0.733$

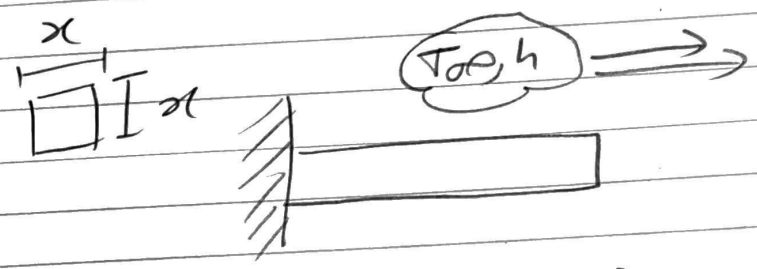
$L_o = 0.0733m$

$\boxed{L_o = 73.3mm}$

min L_o for $T_o \leq 100°C$ is 73.3mm

3. Known: $k = 400 W/m \cdot K$ width $t = s = 0.01 m$

$\eta_f = 0.6$ $h = 100 W/m^2 \cdot K$



Find: a) Fin length b) R_f & ε_f

$A_c = s^2 = 10^{-4} m^2$ $m = \sqrt{\frac{hP}{kA_c}} = 10 m^{-1}$

$P = 0.04 m$

a) $\eta_f = \frac{\tanh(mL_c)}{m(L_c)}$ $L_c = L + \frac{t}{2}$

$(mL_c)0.6 = (\tanh(mL_c))$

$x = 1.51$ Solved (using Desmos)

$mL_c = 1.51$

$L_c = 0.151 m$

$L = L_c - \frac{t}{2}$

$\boxed{L = 0.146 m = 146 mm}$

3.b) ~~$A_f = PL_c$~~

~~$q_f = \eta_f A_f$~~

~~$q_f = \eta_f h A_f \theta_b$~~

$$A_f = L_c P = 6.04 \times 10^{-3}\ m^2$$

$$R_f = \frac{1}{\eta_f h A_f}$$

$$\boxed{R_f = 2.76\ K/W}$$

$$q_f = \eta_f h A_f \theta_b \qquad \varepsilon_f = \frac{q_f}{h A_c \theta_b}$$

$$\therefore \varepsilon_f = \frac{\eta_f A_f}{A_c}$$

$$\boxed{\varepsilon_f = 36.24}$$